MATH 241: Calculus III

Fall 2026

Lecture 25: Surface Integrals

Background & Motivation

We've integrated functions along curves (line integrals) and over flat regions (double integrals). Today we integrate over curved surfaces in 3D. Surface integrals compute mass of a shell, and flux integrals measure how much a vector field flows through a surface. These are the ingredients we need for Stokes' and the Divergence Theorem.

1. Parametric Surfaces

Definition 1: Parametric Surface

A **parametric surface** S is the image of a continuous map $\mathbf{r} : D \to \mathbb{R}^3$, where $D \subseteq \mathbb{R}^2$. Each pair $(u, v) \in D$ is sent to a point on S — think of latitude and longitude on a globe.

$$\mathbf{r}(u, v) = \langle x(u, v),\ y(u, v),\ z(u, v)\rangle, \qquad (u, v) \in D.$$

– **Tangent vectors:** $\mathbf{r}_u = \dfrac{\partial \mathbf{r}}{\partial u}, \quad \mathbf{r}_v = \dfrac{\partial \mathbf{r}}{\partial v}$
– **Normal vector:** $\mathbf{n} = \mathbf{r}_u \times \mathbf{r}_v$ (perpendicular to S; nonzero where S is smooth)
– **Surface area element:** $dS = \|\mathbf{r}_u \times \mathbf{r}_v\|\, du\, dv$

Common Parameterizations

Surface	Parameterization $\mathbf{r}(u,v)$	Domain	$dS = \|\mathbf{r}_u \times \mathbf{r}_v\|\, du\, dv$
Graph $z = g(x, y)$	$\langle x,\ y,\ g(x, y)\rangle$	$(x, y) \in D$	$\sqrt{1 + g_x^2 + g_y^2}\, dA$
Sphere $x^2 + y^2 + z^2 = a^2$	$\langle a \sin\phi \cos\theta,\ a \sin\phi \sin\theta,\ a \cos\phi\rangle$	$\theta \in [0, 2\pi],\ \phi \in [0, \pi]$	$a^2 \sin\phi\, d\phi\, d\theta$
Cylinder $x^2 + y^2 = R^2$	$\langle R\cos\theta,\ R\sin\theta,\ z\rangle$	$\theta \in [0, 2\pi],\ z \in [c, d]$	$R\, d\theta\, dz$
Cone $z = \sqrt{x^2 + y^2}$	$\langle r\cos\theta,\ r\sin\theta,\ r\rangle$	$\theta \in [0, 2\pi],\ r \in [0, h]$	$r\sqrt{2}\, dr\, d\theta$
Plane thru P_0, dirs $\mathbf{a}, \mathbf{b}$	$P_0 + s\mathbf{a} + t\mathbf{b}$	$(s, t) \in D$	$\|\mathbf{a} \times \mathbf{b}\|\, ds\, dt$
Rev. of $y = f(x)$ about x-axis	$\langle x,\ f(x)\cos\theta,\ f(x)\sin\theta\rangle$	$x \in [a, b],\ \theta \in [0, 2\pi]$	$\lvert f(x)\rvert\sqrt{1 + f'(x)^2}\, dx\, d\theta$

Special Case: $z = g(x, y)$

$$dS = \sqrt{1 + g_x^2 + g_y^2}\, dA$$

$$\mathbf{n}\, dS = \langle -g_x,\ -g_y,\ 1\rangle\, dA \quad \text{(upward-pointing normal)}$$

For downward normal, negate: $\langle g_x,\ g_y,\ -1\rangle\, dA$.

2. Scalar Surface Integrals

Definition 2: Surface Integral of a Scalar Function

The **surface integral** $\iint_S f\,dS$ generalizes the double integral from a flat 2D region to a curved surface in $\mathbb{R}^3$. It sums values of f over S weighted by the true area element dS.

$$\iint_S f\,dS = \iint_D f(\mathbf{r}(u,v))\,\|\mathbf{r}_u \times \mathbf{r}_v\|\,du\,dv$$

For $z = g(x,y)$:

$$\iint_S f\,dS = \iint_D f(x,y,g(x,y))\,\sqrt{1+g_x^2+g_y^2}\,dA$$

– **Applications:** mass ($f = \rho$), average value on a surface

Special Case: $f \equiv 1$ gives Surface Area

Setting $f = 1$ in the surface integral turns it into the **surface area** of S:

$$\text{Area}(S) \;=\; \iint_S 1\,dS \;=\; \iint_D \|\mathbf{r}_u \times \mathbf{r}_v\|\,du\,dv$$

For a graph $z = g(x,y)$ over a region D in the xy-plane:

$$\text{Area}(S) \;=\; \iint_D \sqrt{1+g_x^2+g_y^2}\,dA$$

Example 1: Mass of a Hemispherical Shell

Find the mass of the upper hemisphere $x^2+y^2+z^2 = 9$, $z \geq 0$, with density $\rho = z$.

1. **Parameterize:** $\mathbf{r}(\theta,\phi) = \langle 3\sin\phi\cos\theta,\ 3\sin\phi\sin\theta,\ 3\cos\phi\rangle$
   $\theta \in [0,2\pi]$, $\phi \in [0,\pi/2]$; $\|\mathbf{r}_\theta \times \mathbf{r}_\phi\| = 9\sin\phi$
2. **Set up:** $\rho = z = 3\cos\phi$
   $$M = \int_0^{2\pi}\int_0^{\pi/2} (3\cos\phi)(9\sin\phi)\,d\phi\,d\theta$$
3. **Evaluate:** $= 27\int_0^{2\pi} d\theta \int_0^{\pi/2} \sin\phi\cos\phi\,d\phi = 27\cdot 2\pi\cdot\dfrac{1}{2} = \boxed{27\pi}$

3. Flux Integrals

Definition 3: Flux of a Vector Field Through a Surface

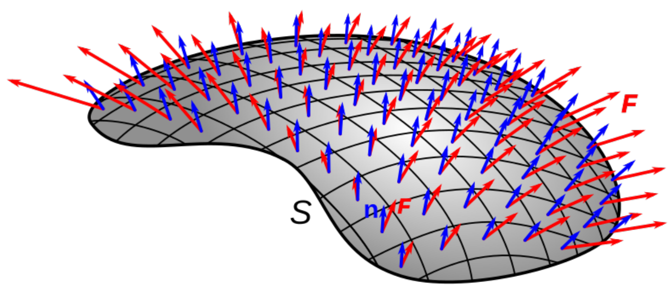
The **flux** of a vector field $\mathbf{F}$ across an oriented surface S measures the net rate at which $\mathbf{F}$ crosses S in the direction of the chosen unit normal $\hat{\mathbf{n}}$. If $\mathbf{F}$ is a fluid velocity field, the flux is the volume of fluid passing through S per unit time.

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D \mathbf{F}(\mathbf{r}(u,v)) \cdot (\mathbf{r}_u \times \mathbf{r}_v)\, du\, dv$$

For $z = g(x,y)$ with **upward** normal:

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D (-P\, g_x - Q\, g_y + R)\, dA$$

– **Interpretation: net flow of F through the surface per unit time**
– Positive flux = net flow in direction of chosen normal
– **Orientation matters:** flipping normal negates the flux



Surface S with normal vectors **n** (blue) and vector field **F** (red). The flux integral measures how much **F** flows through S.

Example 2: Flux Through a Paraboloid

Find the flux of $\mathbf{F} = \langle y, x, z \rangle$ through the paraboloid $z = 1 - x^2 - y^2$, $z \geq 0$, with upward normal.

1. **Set up:** $g(x,y) = 1 - x^2 - y^2$, so $g_x = -2x$, $g_y = -2y$
   $P = y$, $Q = x$, $R = z = 1 - x^2 - y^2$
   Domain D: $x^2 + y^2 \leq 1$ (disk of radius 1)
2. **Apply flux formula:**

$$\begin{aligned} \text{Flux} &= \iint_D (-y(-2x) - x(-2y) + (1 - x^2 - y^2))\, dA \\ &= \iint_D (2xy + 2xy + 1 - x^2 - y^2)\, dA \\ &= \iint_D (4xy + 1 - x^2 - y^2)\, dA \end{aligned}$$

3. **Polar coordinates:** $\displaystyle\int_0^{2\pi} \int_0^1 (4r^2 \cos\theta \sin\theta + 1 - r^2)\, r\, dr\, d\theta$
   The $4r^2 \cos\theta \sin\theta$ term integrates to 0 over $[0, 2\pi]$.
   $\displaystyle= \int_0^{2\pi} \int_0^1 (r - r^3)\, dr\, d\theta = 2\pi \left[\frac{r^2}{2} - \frac{r^4}{4}\right]_0^1 = 2\pi \cdot \frac{1}{4} = \boxed{\frac{\pi}{2}}$

Example 3: Flux Through a Sphere

Find the outward flux of $\mathbf{F} = \langle x,\, y,\, z\rangle$ through the sphere $x^2 + y^2 + z^2 = a^2$.

1. **Outward unit normal:** On the sphere, $\hat{\mathbf{n}} = \dfrac{1}{a}\langle x, y, z\rangle$
2. **Dot product:** $\mathbf{F} \cdot \hat{\mathbf{n}} = \dfrac{x^2 + y^2 + z^2}{a} = \dfrac{a^2}{a} = a$
3. **Integrate:** $\displaystyle\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S a\, dS = a \cdot 4\pi a^2 = \boxed{4\pi a^3}$

Practice Problems

Problem 1. Find the surface area of the plane $z = 3 - x - y$ over the triangle $0 \le x \le 1$, $0 \le y \le 1 - x$.

Answer: $\dfrac{\sqrt{3}}{2}$

Problem 2. Find the flux of $\mathbf{F} = \langle 0, 0, 1\rangle$ upward through $z = 1 - x^2 - y^2$ above $z = 0$.

Answer: π

Problem 3. Compute $\displaystyle\iint_S z\, dS$ where S is the cone $z = \sqrt{x^2 + y^2}$, $0 \le z \le 2$.

(*Hint:* $g_x = x/\sqrt{x^2 + y^2}$, $g_y = y/\sqrt{x^2 + y^2}$, so $\sqrt{1 + g_x^2 + g_y^2} = \sqrt{2}$.)

Answer: $\sqrt{2}\int_0^{2\pi}\int_0^2 r \cdot r\, dr\, d\theta = \sqrt{2} \cdot 2\pi \cdot \frac{8}{3} = \frac{16\sqrt{2}\,\pi}{3}$

Problem 4. Find the flux of $\mathbf{F} = \left\langle x^2, y^2, z^2 \right\rangle$ outward through the unit sphere $x^2 + y^2 + z^2 = 1$. (*Hint: on the unit sphere the outward unit normal is* $\hat{\mathbf{n}} = \langle x, y, z \rangle$*, so* $\mathbf{F} \cdot \hat{\mathbf{n}} = x^3 + y^3 + z^3$*. Use symmetry.*)

Answer: 0 $\quad$ ($\mathbf{F} \cdot \hat{\mathbf{n}} = x^3 + y^3 + z^3$; each term is odd in one variable on a symmetric sphere)

Problem 5. Find the flux of $\mathbf{F} = \langle z, x, y \rangle$ upward through the triangle cut from the plane $x + y + z = 1$ in the first octant.

Answer: $-Pg_x - Qg_y + R = -z(-1) - x(-1) + y = z + x + y = 1;$ $\quad$ Flux $= \iint_D 1\, dA = \frac{1}{2}$